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Short Communication

On the three explosion limits of an H_2–O_2 system and their relationships to ignition delay

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ABSTRACT

We propose a unified model for the three branches of the explosion limits of an H_2–O_2 mixture. By using different expressions for the ignition delay time at each limit, and by utilizing the Le-Chatelier rule, we obtain an analytical expression for the ignition delay time of the mixture. We show that by solving the time delay equation for a selected typical ignition delay time we obtain a single expression for the Z-shape explosion limits.

Explosion limits are the pressure-temperature boundaries for a specific fuel and oxidizer mixture ratio that separate the regions of slow and fast reactions [1]. When a given mixture is heated at a specific pressure, the reaction rate increases with the temperature and gradually turns into a fast reaction. Time delay of this mixture is shortened accordingly. Here, we propose a possible relationship between the time delay and the upper, intermediate and lower explosion limits of an H_2–O_2 system.

There has been a significant amount of research in recent years in the field of alternative fuels, motivated by both environmental and economical aspects. As hydrogen is both clean and efficient fuel, with relatively high enthalpy of combustion (40 W h kg^{-1} compared to common hydrocarbon fuels with 12 W h kg^{-1}), it has the potential to be used in many applications as a future fuel, whether in combustion [2,3] or in fuel cells [4]. However, explosion of hydrogen can occur relatively at low temperatures, especially if stored at high pressures [5]. That is why the study of the explosion limits of hydrogen is of utmost importance to any future development of hydrogen-based energy system, whether it is a low pressure fuel cell, or a high pressure storage tank.

The explosion limits curve for H_2–O_2 system is characterized by three different ignition mechanisms (appearing as three branches in the p-T plane) [1]. The lower explosion limit (lower pressure branch) is determined by a balance between chain-carriers production mainly through gas-phase reactions and chain-carriers removal mainly on the surface (wall effects). As the pressure is raised, the chain-carriers production increases to a point at which surface destruction is no longer sufficient to prevent a branching explosion. The

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intermediate explosion limit (medium pressure branch) is explained by a balance between chain-carriers production through gas-phase reactions and chain-carriers removal through three-body gas-phase reactions. As the pressure is raised further, the latter become more dominant and above a specific pressure the rate of chain-carriers removal exceeds the rate of the chain-carriers production. The upper explosion limit (upper pressure branch) is attributed to the higher gas-phase density that promotes reactions that enhance chain-carriers production. Above a specific pressure there is a rapid increase in the number of chain-carriers species. We note that Wang and Law [6] have shown and suggested that a system explosion can still occur also around the upper explosion limit by accounting for the branched-chain alone without heat evolvement.

In an earlier work [7] we examined the ignition temperatures, considering the upper and intermediate explosion limits only, and showed that a unified explosion limit is obtained in the form of a single analytical expression.

In the present note we consider the three simultaneous mechanisms that are dominated by the gas-phase and surface reactions, namely the upper, the intermediate and the lower explosion limits. We show that by employing the Le-Chatelier rule to evaluate the overall time delay (while considering the three-mechanisms), a simple relationship between the overall time delay and the explosion limits of an H_2–O_2 system can be derived. For the lower and intermediate branches we use the following reactions' scheme [8]:

$$\begin{array}{lll}
R0 & H_2+O_2 \rightarrow H+HO_2-55kcal & \text{Initiation} \\
R1 & H_2+OH \rightarrow H_2O+H+14kcal & \\
R2 & H+O_2 \rightarrow OH+O-15kcal & \text{Branched chain propagation} \\
R3 & H_2+O \rightarrow OH+H+0kcal & \\
R6 & H+O_2+M \rightarrow HO_2+M & \text{Chain termination} \\
R1' & OH+wall \rightarrow destruction & \\
R2' & H+wall \rightarrow destruction & \text{Wall termination} \\
R3' & O+wall \rightarrow destruction & \\
R6' & HO_2+wall \rightarrow destruction &
\end{array} \tag{1}$$

The production rate of the H species is:

$$\frac{d[H]}{dt} = n_0 + a_1[OH] - \left(a_2 + a_2' + a_6\right)[H] + a_3[O] \tag{2}$$

where $n_0 = k_0[H_2][O_2]$ and a_i is the modified reaction rate constant of the *i-th* reaction. The production rates of the other relevant species are:

$$\frac{d[OH]}{dt} = -\left(a_1 + a_1'\right)[OH] + a_2[H] + a_3[O] \tag{3}$$

$$\frac{d[O]}{dt} = a_2[H] - \left(a_3 + a_3'\right)[O] \tag{4}$$

$$\frac{d[HO_2]}{dt} = a_6[H] - a_6'[HO_2] \tag{5}$$

While the overall rate, defined by the oxygen consumption is:

$$-\frac{d[O_2]}{dt} = \frac{d[\Delta O_2]}{dt} = n_0 + (a_2 + a_6)[H] \tag{6}$$

where $\Delta[O_2] = [O_2]_{t=0} - [O_2]_t$. Here:

$$\begin{aligned}
a_1 &= k_1[H_2] = A_1 T^{n_1} \exp\left(-\frac{E_1}{RT}\right)[H_2] \\
a_2 &= k_2[O_2] = A_2 T^{n_2} \exp\left(-\frac{E_2}{RT}\right)[O_2] \\
a_3 &= k_3[H_2] = A_3 T^{n_3} \exp\left(-\frac{E_3}{RT}\right)[H_2] \\
a_6 &= k_6[O_2][M] = A_6 T^{n_6} \exp\left(-\frac{E_6}{RT}\right)[O_2][M] \\
a_1' &= \frac{u_{OH}\varepsilon_{OH}}{4}\left(\frac{S}{V}\right) \\
a_2' &= \frac{u_H\varepsilon_H}{4}\left(\frac{S}{V}\right) \\
a_3' &= \frac{u_O\varepsilon_O}{4}\left(\frac{S}{V}\right) \\
a_6' &= \frac{u_{HO_2}\varepsilon_{HO_2}}{4}\left(\frac{S}{V}\right)
\end{aligned} \tag{7}$$

where k_i is the rate constant of the *i-th* reaction, with A_i, n_i, E_i being the Arrhenius coefficients. For the wall reactions, u_i is the thermal velocity, ε_i the surface reactivity and $\left(\frac{S}{V}\right)$ the surface-to-volume ratio of the vessel.

Semenov [8] assumed that the rate of H formation dominates the ignition success and thus at ignition, $\frac{d[O]}{dt} = 0$ and $\frac{d[OH]}{dt} = 0$. Semenov further assumed that during the ignition delay time the concentrations of oxygen and hydrogen are fairly constant, and thus the temperature and pressure do not vary significantly. Under these conditions, $\frac{d[H]}{dt}$ (Equation (2)) may be integrated to yield: $[H] = \frac{n_0}{|\phi|}[\exp(-|\phi|t) - 1]$ where:

$$\phi = a_1 a_2 \frac{1 + \frac{a_3}{(a_3 + a_3')}}{(a_1 + a_1')} - \left(a_2 + a_2' + a_6\right) + \frac{a_2 a_3}{(a_3 + a_3')} \tag{8}$$

Following Semenov judicious assumptions for a stoichiometric mixture of H_2 and O_2, these equations may be solved to yield the lower (*lo*) and intermediate (*im*) pressures at which ignition occurs: $p_{lo} = \frac{a_2'}{2k_2\chi_{O_2}N}$ and $p_{im} = \frac{2k_2}{k_6 N}$, where χ_{O_2} is the initial oxygen mole fraction and N is the number of molecules per unit volume.

On the basis of a fairly big volume of experimental data, Semenov suggested that the product of the ignition time delay of the mixture at low pressures (in the vicinity of the lower and intermediate explosion limits), τ_{lo-im}, and the parameter ϕ, is constant ($\tau_{lo-im}\phi = const$). Following Semenov's term for ϕ (Equations (7) and (8)), assuming $\frac{a'_1}{a_1} < 1$ and $\frac{a'_3}{a_3} < 1$, we can show that:

$$\phi = f(T)\left(p - p_{lo}\right)\left(p_{im} - p\right) \tag{9}$$

While realizing that the parameter ϕ is a complicated function of the pressure and temperature, we propose here a general form of the type of:

$$\tau_{lo-im} = \left(C_{lo}p^{n_{lo}} + C_{im}p^{n_{im}}\right)C_T \exp\left(\frac{E_T}{T}\right) \tag{10}$$

The time delay is a product of two independent terms; one being pressure dependent and the other temperature dependent. The pressure dependent term is composed of two terms that characterize the lower and the intermediate explosion limits respectively. The six constants, C_{lo}, C_{im}, C_T, n_{lo}, n_{im}, and E_T, are rational calibration constants.

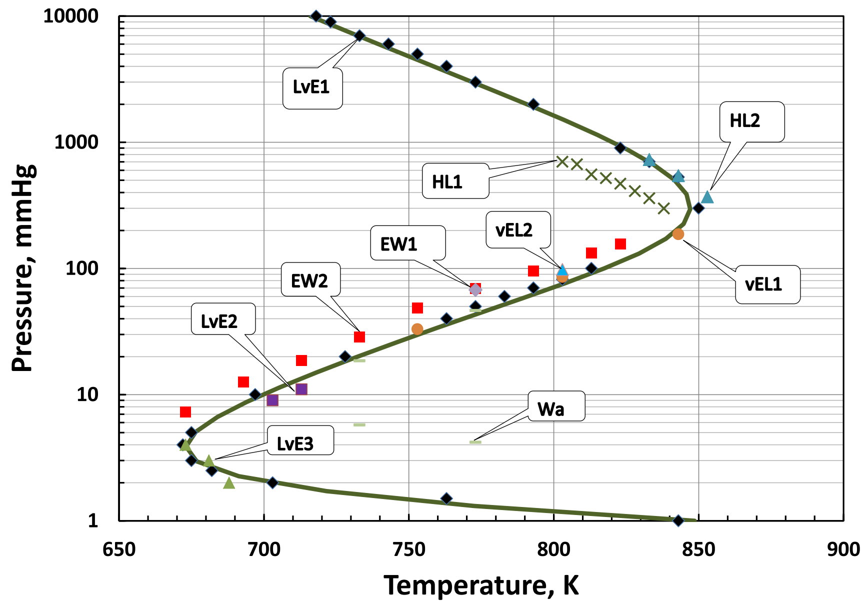

Fig. 1 – Our theoretical results (solid curve) compared to various experimental studies. Experimental data is for 7.3–7.4 cm diameter spherical Pyrex vessels with different coatings as follows: LvE1: KCl coating, summary of experiments and lines extrapolation (lowest and highest pressures are extrapolated) [12], LvE2: light KCl coating [12], LvE3: heavy KCl coating [12], HL1: light KCl coating [14], HL2: heavy KCl coating [14], vEL1: KCl coating [15], vEL2: without coating [15], vEL3: salt coating [15], Wa: KCl coating [16], EW1: B2O3 coating [17], EW2: Boric Acid coating [17].

For the upper branch we use the well accepted general correlation between the time delay and the thermodynamic conditions of the mixture (e.g. Snyder et al. [9]). It is however important to note that the complementary chain-carriers theory suggests that self ignition occurs when the rate of chain-carriers production exceeds the rate of chain-carriers removal. Azatyan [10] showed that when the chain-carries principle is introduced into the classical thermal explosion criterion of Frank-Kamenetskii, the upper explosion limit may take the form of:

$$\tau_{up} = C_{up} p^{n_{up}} \exp\left(\frac{E_{up}}{RT}\right) \tag{11}$$

where C_{up}, n_{up}, and E_{up}, are constants to be calibrated. This correlation suggests that the mixture will explode at any given pressure and temperature while the only distinctive parameter that may span over a number of order of magnitudes is the time delay. This is clearly in full agreement with the explosion limit that is defined as the pressure-temperature boundaries that separate the regions of slow and fast reactions.

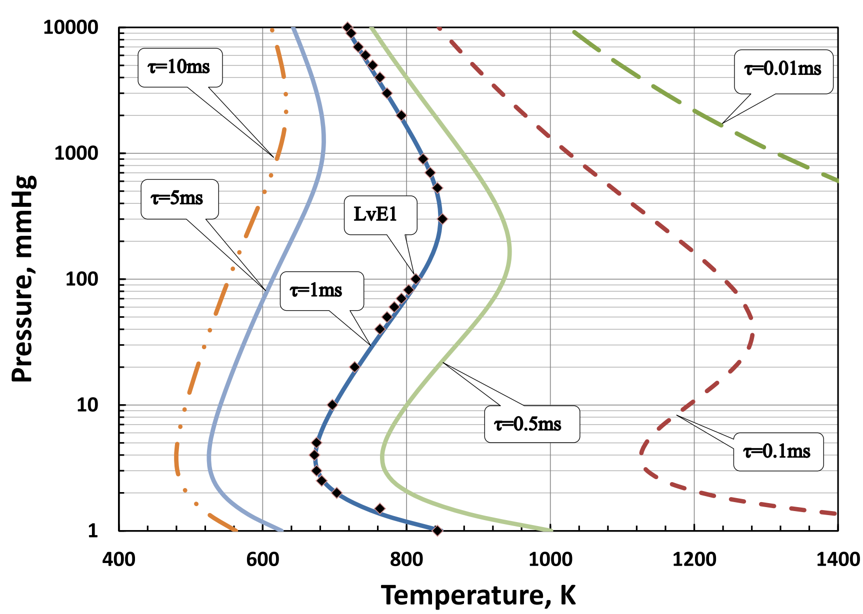

Fig. 2 – Iso-delay time contours for different values of ignition delay time (rhombi shapes are experimental data of LvE1 [12]).

The ignition delay time of the mixture under any initial conditions is a result of the three mechanisms occurring simultaneously. Each mechanism dominates the time delay in its region but two mechanisms may have comparable effects in particular in the vicinity of the two turnover points. The resultant time delay in any region may therefore be evaluated from the Le-Chatelier rule (Khitrin [11]):

$$\frac{1}{\tau} = \frac{1}{\tau_{up}} + \frac{1}{\tau_{\tau_{lo-im}}} \tag{12}$$

We postulate here that the ignition limit curve (the complete "peninsula" curve for the three explosion limits) may be represented by an arbitrary pre-specified value of the time delay (an "iso-time delay" curve). The sensitivity of the results to the arbitrary pre-specified value will be evaluated below.

We select a representative value for the time delay of 1 ms. For best fitting the lower and the intermediate limits branches to the experimental values of Lewis and von Elbe [12] and Mass and Warnatz [13], while setting the time ignition of each to 1 ms, we obtain: $C_{lo} = 2.686$, $C_{im} = 0.51$, $C_T = 3.37 \cdot 10^{-6}$, $n_{lo} = -2.2$, $n_{im} = 0.372$, and $E_T = 7.69$. By fitting the upper limit branch to the experimental values, we obtain: $C_{up} = 30 \cdot 10^{-7}$, $n_{up} = -1.05$, and $E_{up} = 22.38$. Equations (10)–(12) thus define the contours of iso-time delays. Fig. 1 shows our results in comparison to various experimental results. We note that while the constants were selected to fit best the experimental data, the relationship types are solidly based on the chemistry and physics fundamentals of the problem. Fig. 2 shows how sensitive are the iso-time delay contours to the value of the arbitrary selected value of the time delay. A change in value of one order of magnitude affects the ignition temperature by 200–400°C, thus demonstrating the concept of the explosion limits definition, i.e., as the pressure-temperature boundaries that separate the regions of slow and fast reactions.

In summary, we show that the ignition limit curve (the complete "peninsula" curve for the three explosion limits) may be represented by an arbitrary pre-specified value of the time delay (an "iso-time delay" curve). Equations (10)–(12) above that are solidly based on the chemistry and physics fundamentals of the problem, define the contours of the iso-time delay, while the constants are selected to fit best the experimental observations.

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